

TLS + FABP5 Validation Memo

Status update for the SAGE Figure 3 rebuttal work. Prepared by Tanvir. July 27, 2026.

Summary

The team asked me to test one claim from the SAGE manuscript. The claim is this: patients with high FABP5 gene expression and few tertiary lymphoid structures (TLS) have worse survival than patients with low FABP5 and many TLS. The manuscript reports this result in TCGA-BLCA, a public bladder cancer dataset, with a hazard ratio of 2.34 and a p-value of 0.013.

I tested this claim on two other cancer types: colon (TCGA-COAD) and lung squamous cell carcinoma (TCGA-LUSC). Neither test showed a significant result. I also tried four different ways to rebuild the original BLCA result from public data. None of the four matched the manuscript's numbers. I traced the code behind the manuscript's method and found the real algorithm is more complex than a simple split on FABP5 and TLS. The exact per-patient list behind the manuscript's Figure 3 is still missing, and finding it is the one open item left.

What is still open

- The exact patient list behind the manuscript's BLCA result. Someone named Jinchu is checking this now.
- A final decision from Sahar and the team on whether the colon and lung results count as a full null finding.

1. Where the work started

The first request came from Sahar and Sandeep. They sent two files for a bladder cancer dataset called TRRC: a clinical file with 93 patients, and a TLS feature file with 154 rows. The goal was to check if these files could support a TLS density calculation and a survival analysis.

I checked both files and found two problems.

- No FABP5 gene expression data existed anywhere in either file.
- The TLS-related columns (TLS, TSR, TLSTSR) had no documented formula, and were not built for a density calculation. They looked like a separate, older scoring method with an unknown source.

Because of this, the team moved the validation target from TRRC to a public dataset with both FABP5 and TLS data available: TCGA-COAD, the colon cancer arm of The Cancer Genome Atlas.

2. Colon (TCGA-COAD)

Sandeep supplied FABP5 expression and survival data for 265 colon patients. Sahar supplied 120 TLS mask images. Sandeep supplied a link to public tumor-boundary annotations to compute tumor area.

2.1 Data quality problem

100 of the 120 TLS masks were completely blank. Only 19 of 114 matched patients had any detectable TLS at all. This is a severe data limitation. Sahar and Sandeep confirmed these masks came from a 1:4 pixel downsampling relative to the full-resolution slide, which let us compute a real area once the masks did show something.

2.2 Results

Analysis	Groups compared	n (events)	Hazard Ratio [95% CI]	p-value
Path B: TLS detected vs. not (log-rank)	Detected (19) vs. not (95)	114 (32)	HR=1.10 [0.45, 2.69]	0.827
Path B: FABP5 high vs. low (median split)	High (56) vs. low (58)	114 (32)	HR=1.18 [0.59, 2.36]	0.648
Path C: continuous TLS density (% of tumor area)	Continuous, all patients	114 (32)	HR=0.68 per % point	0.742
Corner comparison, mirroring the manuscript	High FABP5 (colon TLS) vs. Low FABP5 (colon TLS)	55 (19)	HR=0.68 [0.22, 2.08]	0.500
Corner comparison, corrected to use BLCA	High FABP5 (BLCA median) vs. Low FABP5 (BLCA median)	55 (19)	HR=0.68 [0.22, 2.08]	0.500

None of the five colon analyses reached significance. The point estimate direction was inconsistent across analyses, which is expected with this few detected-TLS patients (9 to 19 depending on the test). This is a genuine null result, not a formatting problem.

2.3 A methods correction made along the way

The first corner comparison used colon's own FABP5 median as the high/low cutoff. This is not a correct external validation test. A true external test applies the cutoff learned on the original dataset (BLCA) to the new dataset (colon), without re-deriving a new cutoff from colon itself. I pulled real FABP5 values for TCGA-BLCA from cBioPortal, computed BLCA's own median (63.10 TPM), and reran the colon comparison using that fixed external cutoff. The result stayed null.

3. Pivot to lung (TCGA-LUSC)

Because the colon result was null, and because the colon TLS mask data quality was poor, the team decided to test the same claim on a third, independent cancer type. Sandeep first named this cohort TCGA-LUAD (adenocarcinoma), then corrected it to TCGA-LUSC (squamous cell carcinoma). All prior work done under the LUAD name was replaced with the correct LUSC data.

Sandeep sent three links: TLS annotations (HookNet-TLS, a public dataset from Zenodo), tumor-region annotations (a second Zenodo dataset), and a cBioPortal study ID for FABP5 and survival data. All three were real, working, public sources.

3.1 A pixel-scale problem, and how it was solved

The TLS annotations are fixed at about 0.5 micrometers per pixel for every slide. The tumor annotations are in each slide's own native pixel scale, which is not fixed. To compute a real density (TLS area over tumor area), both areas need to be in the same unit.

I recovered each slide's native scale directly from the TLS dataset itself, by comparing its two bundled coordinate systems against each other. 371 of 375 slides resolved cleanly to one of two standard scanner values: 0.25 or 0.5 micrometers per pixel. This matched, and slightly refined, a scaling note Sandeep sent independently.

3.2 Two tumor-area sources were used

Source	Slides covered	Final cohort (n)	TLS detected
Crowd-sourced Zenodo tumor masks (first pass)	323	203	189 (93%)
Sandeep's in-house pathologist tumor annotations (second pass, more precise)	497	331	305 (92%)

The pathologist file is the more reliable source. It has precise, per-slide pixel scale and removes overlapping polygons. All final conclusions below rely on this 331-patient version.

3.3 Results, all versions

Analysis	Cohort	n (high/low)	Hazard Ratio [95% CI]	p-value
Step 1: BLCA-derived thresholds (both FABP5 and density)	Zenodo tumor mask (n=203)	107 (23/84)	HR=1.04 [0.55, 1.97]	0.906
Step 2: LUSC's own medians (both FABP5 and density)	Zenodo tumor mask (n=203)	104 (52/52)	HR=1.16 [0.68, 1.96]	0.592
Continuous, no threshold, density and FABP5 combined	Zenodo tumor mask (n=203)	203	density HR=0.77/SD	0.053
Step 1: BLCA-derived thresholds (both FABP5 and density)	Pathologist tumor area (n=331)	175 (25/150)	HR=1.23 [0.70, 2.18]	0.475
Step 2: LUSC's own medians (both FABP5 and density)	Pathologist tumor area (n=331)	164 (82/82)	HR=1.17 [0.74, 1.85]	0.510
Continuous, no threshold, density and FABP5 combined	Pathologist tumor area (n=331)	331	density HR=0.95/SD	0.611
Univariate, TLS density alone, median split, full follow-up	Pathologist tumor area (n=331)	331 (165/166)	HR=0.76 [0.55, 1.04]	0.081
Univariate, FABP5 alone, median split, full follow-up	Pathologist tumor area (n=331)	331 (165/166)	HR=0.88 [0.65, 1.21]	0.440
Univariate, TLS density alone, median split, censored at 60 months	Pathologist tumor area (n=331)	331 (165/166)	HR=0.82 [0.58, 1.14]	0.239
Univariate, FABP5 alone, median split, censored at 60 months	Pathologist tumor area (n=331)	331 (165/166)	HR=0.80 [0.57, 1.12]	0.187
Continuous, density + FABP5 combined, censored at 60 months	Pathologist tumor area (n=331)	331	density HR=0.99/SD	0.901

One result came close to a trend: TLS density alone, full follow-up, $p = 0.081$. This did not hold up under two independent checks. It weakened, not strengthened, when tumor area came from the more precise pathologist source (versus the smaller, cruder Zenodo cohort). It weakened further when follow-up was capped at 60 months, as requested, dropping to $p = 0.239$. A real effect should get clearer with better data, not weaker. This pattern points to noise, not a missed signal.

Combining continuous TLS density with continuous FABP5 in one model, with or without an interaction term, did not improve the result at any point. Every combined model stayed non-significant.

4. The BLCA patient list request

Separately, Sandeep asked for the list of patients from the manuscript's original TCGA-BLCA result: patient ID in one column, and a risk score of 1 (high risk) or 0 (low risk) in a second column. This list was never given to this analysis at the start. Two paths were tried to produce it.

4.1 Rebuilding it from public data

The TLS annotation dataset Sandeep sent for lung also turned out to include real annotations for TCGA-BLCA (345 slides). Combined with real FABP5 values pulled from cBioPortal, this let me try four different ways to rebuild the manuscript's result.

Method	TLS measure used	Corner n	Hazard Ratio	p-value
Manuscript (Figure 3), for reference	Not disclosed	Not disclosed	2.34 [1.25, 4.40]	0.013
Rebuild A	Count of TLS-labeled polygons per slide	149	1.19 [0.73, 1.94]	0.480
Rebuild B	Count of germinal-center polygons per slide	149	1.56	0.109
Rebuild C	TLS area over real tumor area (same method as lung)	130	1.49 [0.88, 2.53]	0.138

None of the four rebuilds matched the manuscript's number. This means the manuscript's exact method for measuring TLS abundance is not a simple polygon count or a simple area ratio.

4.2 Tracing the actual code

Jinchu shared four anonymized code repositories tied to the manuscript: a hypothesis-generation agent, a tools-orchestration system, a data-discovery agent, and a coding agent. I read through all four directly, file by file, rather than only the summary pages.

- The coding agent repository has 6 files total and zero data files of any kind. Its own documentation says it only executes what an upstream pipeline hands it. It does not store data by design.
- The tools-orchestration repository contains the real survival analysis tool. Its method is a bootstrap CoxNet (elastic-net Cox regression) combined with automated feature selection across many bootstrap splits, not a simple two-variable median split. The risk cutoff is set from the training set's score distribution, then applied to a separate holdout set.
- That same tool depends on a separate tool called `combine_features` to build its input file. Neither tool's actual input or output data is stored in the repository, only the code that would produce it.
- All three repositories checked have a `.gitignore` file that excludes run-output folders (named `workspaces` or `runs`) by design. This means the real per-patient result was very likely never saved into any of these code repositories, at any point in time.

Conclusion: the exact list Sandeep wants cannot come from these repositories. It can only come from wherever the actual output folder was saved when someone ran this code against real BLCA data. Jinchu is checking on this now.

5. Data and files produced

All files below are on this machine, under Downloads unless noted.

File / folder	What it contains
TLS_FABP5_Survival_PathB/	Colon presence/absence survival analysis, PDF and plots
TLS_Density_PathC/	Colon continuous density survival analysis, PDF and plots
LUSC_Density_Results/	First-pass lung results (Zenodo tumor masks), PDF and plots
LUSC_Pathologist_Results/	Final lung results (pathologist tumor area), PDF, plots, and all KM curves
LUSC_patient_TLS_tumor_area.csv	Per-patient TLS area and tumor area for all 203 lung patients (first pass)
tcga_lusc_tumour_area_mm2.csv	Sandeep's pathologist tumor area file, 511 slides (input, not produced by this work)
TCGA_COAD_selected_cases_FABP5_expression_clinical_cohort	Complete clinical and FABP5 data (input, from Sandeep)

6. What is still open

Item	Owner	Status
Exact BLCA per-patient risk list from the manuscript	Jinchu	Checking for a saved output folder
Team decision on whether colon + lung null results are the final rebuttal answer	Sandeep / Sandeep	Awaiting team review